

09

R.T.F :- Whether the mean ($\bar{x}$) is biased or unbiased

For biased $E[\hat{\mu}] = \mu$

$E[\hat{p}] = E[x_1]$ (Since we have expectation operator, we must consider all possibilities of x_1)

$$= 3$$

Q2)

(ii) x is d dimensional datapoints

(iv) For class 1's i^{th} feature generation of datapoint (say x) is Bernoulli Random Variable with parameter θ_i .

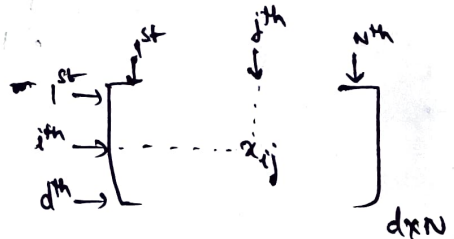
(v). For class 2's " " " parameter ϕ_i .

(vi) Beta prior for feature i is $\text{Beta}(\alpha_i, \beta_i)$

R.T.F. - (i) Posterior $p(w_1|x)$ and $p(w_2|x)$ (where $x = \{x_1, x_2, \dots\}$)

(ii) Decision boundary

Sol :- Let X (data matrix) =



x_{ij} refers to i^{th} feat. of j^{th} datapoint

(Let's consider ~~we~~ we are discriminating N_1 datapoints of w_1 & N_2 points of $w_2 \Rightarrow N_1 + N_2 = N$)

$$P(x_{ij} | \omega_i) = \theta_i^{x_{ij}} (1 - \theta_i)^{1 - x_{ij}}$$

$$P(x_i | \omega_i) = \prod_{j=1}^{N_i} \theta_i^{x_{ij}} (1 - \theta_i)^{1 - x_{ij}}$$

$$P(\omega_i | x_i) = P(x_i | \omega_i) \cdot P(\omega_i) \quad (\text{discarded } P(x_i) \text{ as it doesn't play a role!})$$

$$= \left(\prod_{j=1}^{N_i} \theta_i^{x_{ij}} (1 - \theta_i)^{1 - x_{ij}} \right) \frac{\theta_i^{\alpha_i - 1} (1 - \theta_i)^{\beta_i - 1}}{f(\alpha_i, \beta_i)}$$

$$P(\omega_1 | x) = \prod_{i=1}^d P(\omega_i | x_i) = \prod_{i=1}^d \left(\left(\prod_{j=1}^{N_i} \theta_i^{x_{ij}} (1 - \theta_i)^{1 - x_{ij}} \right) \frac{\theta_i^{\alpha_i - 1} (1 - \theta_i)^{\beta_i - 1}}{f(\alpha_i, \beta_i)} \right)$$

or

$$P(\omega_1 | \{x_1, x_2, \dots, x_{N_1}\})$$

$$= \prod_{i=1}^d \prod_{j=1}^{N_i} \theta_i^{x_{ij}} (1 - \theta_i)^{1 - x_{ij}} \prod_{i=1}^d \frac{\theta_i^{\alpha_i - 1} (1 - \theta_i)^{\beta_i - 1}}{f(\alpha_i, \beta_i)}$$

$$= \prod_{i=1}^d \theta_i^{\sum_j x_{ij}} (1 - \theta_i)^{N_i - \sum_j x_{ij}} \prod_{i=1}^d \frac{\theta_i^{\alpha_i - 1} (1 - \theta_i)^{\beta_i - 1}}{f(\alpha_i, \beta_i)}$$

$$= \prod_{i=1}^d \left(\frac{\theta_i^{\sum_j x_{ij}} (1 - \theta_i)^{N_i - \sum_j x_{ij}} \theta_i^{\alpha_i - 1} (1 - \theta_i)^{\beta_i - 1}}{f(\alpha_i, \beta_i)} \right)$$

$$= \prod_{i=1}^d \left(\frac{\theta_i^{\sum_j x_{ij} + \alpha_i - 1} (1 - \theta_i)^{N_i + \beta_i - \sum_j x_{ij} - 1}}{f(\alpha_i, \beta_i)} \right)$$

$$P(\omega_1 | \{x_1, x_2, \dots, x_{N_1}\}) \text{ or } P(\omega_1 | x) = \prod_{i=1}^d \text{Beta}(\sum_j x_{ij} + \alpha_i, N_i + \beta_i - \sum_j x_{ij})$$

$$\text{Likely } P(\omega_2 | x) = \prod_{i=1}^d \text{Beta}(\sum_j x_{ij} + \alpha_i, N_2 + \beta_i - \sum_j x_{ij}) \text{ or}$$

$$\prod_{i=1}^d \left(\frac{\phi_i^{\sum_j x_{ij} + \alpha_i - 1} (1 - \phi_i)^{N_2 + \beta_i - \sum_j x_{ij} - 1}}{f(\alpha_i, \beta_i)} \right)$$

discriminant

$$g_1(x) - g_2(x) = \ln p(x|\omega_1) - \ln p(x|\omega_2) + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$$p(x|\omega_1) = \prod_{i=1}^d \theta_i^{x_i} (1-\theta_i)^{1-x_i}$$

$$p(x|\omega_2) = \prod_{i=1}^d \phi_i^{x_i} (1-\phi_i)^{1-x_i}$$

$$\ln(p(x|\omega_1)) - \ln(p(x|\omega_2)) = \sum_{i=1}^d (x_i \ln \theta_i + (1-x_i) \ln(1-\theta_i) - x_i \ln \phi_i - (1-x_i) \ln(1-\phi_i))$$

$$= \sum_{i=1}^d x_i (\ln \theta_i - \ln(1-\theta_i) - \ln \phi_i + \ln(1-\phi_i)) + \sum_{i=1}^d (\ln(1-\theta_i) - \ln(1-\phi_i))$$

$$= \sum_{i=1}^d x_i \frac{\ln \theta_i (1-\phi_i)}{\phi_i (1-\theta_i)} + \sum_{i=1}^d \frac{\ln(1-\theta_i)}{(1-\phi_i)}$$

$$g_1(x) - g_2(x) = \sum_{i=1}^d x_i \ln \left(\frac{\theta_i (1-\phi_i)}{\phi_i (1-\theta_i)} \right) + \sum_{i=1}^d \frac{\ln(1-\theta_i)}{(1-\phi_i)} + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

$$\text{thus } w = \left[\ln \left(\frac{\theta_i (1-\phi_i)}{\phi_i (1-\theta_i)} \right) \right]_{i=1, \dots, d}$$

$$\text{and } b = \sum_{i=1}^d \frac{\ln(1-\theta_i)}{(1-\phi_i)} + \ln \frac{p(\omega_1)}{p(\omega_2)}$$

Q3)

Ans) Given :- 1) Mean not removed

2) Proj mean has unit magnitude

3) Maximize the variance of projected samples

R.T.F :- $u \in \mathbb{R}^d$ which satisfies above 3 points in eigenvector equation form

so let data matrix be $X_{d \times N}$

and mean is $\mu = \frac{\sum_{j=1}^N x_j}{N}$

and covariance matrix is $\frac{1}{N-1} (X-\mu)(X-\mu)^T$
(Σ or S)

The projected mean $\mu_{\text{proj}} = u^T \mu$ (a scalar)

The projected mean vect = $\mu_{\text{proj}} \times \hat{u} = \frac{u^T \mu u}{\|u\|_2}$
($\vec{\mu}_{\text{proj}}$)

But the projected mean must be of unit magnitude

$$\|\vec{\mu}_{\text{proj}}\|_2 = \left\| \frac{u^T \mu u}{\|u\|_2} \right\|_2 = \frac{\|u^T \mu u\|_2}{\|u\|_2} = \frac{u^T \mu \|u\|_2}{\|u\|_2} = 1$$

$$\boxed{u^T \mu = 1}$$

The proj variance is $u^T S u$. So, we must
 $\max_u u^T S u$ given or s.t $u^T \mu = 1$

Use Lagrange multiplier to solve $\max u^T S u$ s.t. $u^T \mu = 1 \Rightarrow (u^T \mu)^2 = 1$
 $\Rightarrow u^T \mu \mu^T u = 1$

$$\max_u u^T S u - \lambda (u^T \mu \mu^T u - 1)$$

$$\Rightarrow \nabla_u (u^T S u - \lambda (u^T \mu \mu^T u - 1)) = 0$$

$$\Rightarrow 2 S u - 2 \lambda \mu \mu^T u = 0$$

$$\Rightarrow 2 (S - \lambda \mu \mu^T) u = 0$$

$$\text{Hence } S u = \lambda \mu \mu^T u$$

Q4)

- Ans) Given:-
- 1) Distance between proj mean of binary classes is 1
 - 2) Maximize the scatter of all datapoints (like PCA)
 - 3) Minimize within class
 class scatter
 - 4) $X_1 \in \mathbb{R}^{d \times N_1}$ and $X_2 \in \mathbb{R}^{d \times N_2}$

- R.T.F:-
- 1) Expression for means & covariance scatter
 - 2) Obj. function
 - 3) Find w

$$\mu_{class1} = \sum_{i=1}^{N_1} \frac{x_{1i}}{N_1} \quad \text{||ly} \quad \mu_{class2} = \sum_{i=1}^{N_2} \frac{x_{2i}}{N_2} \quad \left(\begin{array}{l} X_1 = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1N_1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N_11} & x_{N_12} & \dots & x_{N_1N_1} \end{bmatrix} \\ X_2 = \begin{bmatrix} x_{21} & x_{22} & \dots & x_{2N_2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N_21} & x_{N_22} & \dots & x_{N_2N_2} \end{bmatrix} \end{array} \right) \text{||ly}$$

$$\Sigma_{class1} = \frac{1}{N_1} (X_1 - \mu_{class1})(X_1 - \mu_{class1})^T$$

$$\Sigma_{class2} = \frac{1}{N_2} (X_2 - \mu_{class2})(X_2 - \mu_{class2})^T$$

$$S_{class1} = N_1 \times \Sigma_{class1} = (X_1 - \mu_{class1})(X_1 - \mu_{class1})^T$$

$$S_{class2} = N_2 \times \Sigma_{class2} = (X_2 - \mu_{class2})(X_2 - \mu_{class2})^T$$

$$\text{Within the class scatter of only data points (} S_w \text{)} = \sum_{i=1}^2 S_{classi} = \sum_{i=1}^2 (X_i - \mu_{classi})(X_i - \mu_{classi})^T$$

~~$S_{class1} + S_{class2}$~~

$$\text{Within the class scatter (proj datapoints)} = w^T S_w w$$

$$\text{distance between proj mean} = \|w^T \mu_{class1} - w^T \mu_{class2}\|^2$$

$$= w^T (\mu_{class1} - \mu_{class2})(\mu_{class1} - \mu_{class2})^T w$$

$$= w^T m m^T w \quad (m = \mu_{class1} - \mu_{class2})$$

$$w^T m m^T w = 1 \quad (\text{dis. of proj mean} = 1)$$

~~Scatter~~
~~Var. of total data samples S_{TOTAL}~~

For max. total scatter

$$X = [X_1 | X_2] = \begin{bmatrix} \overset{1^{st}}{x_{11}} & \dots & x_{1j} & \dots & \overset{N_1^{th}}{x_{1N_1}} | \overset{(N_1+1)^{th}}{x_{21}} & \dots & \overset{(N_2+N_1)^{th}}{x_{2N_2}} \end{bmatrix} = \begin{bmatrix} x_1 & \dots & \overset{j^{th}}{x_j} & \dots & \overset{2N^{th}}{x_{N'}} \end{bmatrix}$$

where $N' = N_1 + N_2 = 2N$

$$\text{Covariance of } X(\Sigma) = \frac{1}{N'} \sum_{\substack{i=1 \\ x_i \in X}}^{N'} (x_i - \mu_{\text{total}})(x_i - \mu_{\text{total}})^T$$

$$\text{where } \mu_{\text{total}} = \frac{1}{N'} \sum_{\substack{i=1 \\ x_i \in X}}^{N'} x_i = \frac{1}{2N} \left(\sum_{\substack{i=1 \\ x_i \in X_1}}^{N_1} x_i + \sum_{\substack{i=N_1+1 \\ x_i \in X_2}}^{N_2} x_i \right) = \frac{1}{2N} (N\mu_{\text{class1}} + N\mu_{\text{class2}})$$

$$= \frac{\mu_{\text{class1}} + \mu_{\text{class2}}}{2}$$

$$\Sigma = \frac{1}{2N} (X - \mu_{\text{total}})(X - \mu_{\text{total}})^T$$

$$\text{scatter } (S_{\text{total}}) = 2N \times \Sigma = (X - \mu_{\text{total}})(X - \mu_{\text{total}})^T$$

$$\text{proj scatter} = w^T S_{\text{total}} w$$

$$\text{Obj function :- } \max_w \frac{w^T S_{\text{total}} w}{w^T S_w w} \quad \text{s.t. } w^T m m^T w = 1$$

The objective function, after rayleigh ^{quotient} coefficient, becomes

$$\max_y y^T S_T y \quad \text{s.t.} \quad y^T S_w y = 1 \quad \text{and} \quad y^T S_B y = 1. \quad (\text{where } S_B = mm^T \begin{pmatrix} \mu_1 - \mu_2 \\ \times (\mu_1 - \mu_2)^T \end{pmatrix}).$$

By applying langrange multiplier, we get

$$\nabla_y (y^T S_T y - \lambda_1 (y^T S_w y - 1) - \lambda_2 (y^T S_B y - 1)) = 0$$

$$2S_T y - \lambda_1 (2S_w y) - \lambda_2 (2S_B y) = 0$$

$$S_T y = \lambda_1 S_w y + \lambda_2 S_B y$$

Q5)

Ans) Given :- $X = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}_{d \times N}$ (where $d=N=2$)

Take unbiased estimate of covariance matrix

R.T.F :- U (PCA matrix)

$$\Sigma = \frac{1}{N-1} (X - \mu)(X - \mu)^T$$

$$\mu = \frac{1}{N} \sum_{i=1}^N (x_i) = \frac{x_1 + x_2}{2} = \frac{\begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix}}{2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{aligned}
 \Sigma &= (X - \mu)(X - \mu)^T \\
 &= \left(\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \left(\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)^T \\
 &= \left(\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \right) \left(\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \right)^T \\
 &= \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}
 \end{aligned}$$

The PC are the eigenvectors of covariance matrix. Here, u is PC.

$$(\Sigma - \lambda I)u = 0$$

$$|\Sigma - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 - 4 = 0$$

$$(\lambda^2 - 4\lambda + 4) - 4 = 0$$

$$\lambda(\lambda - 4) = 0$$

$$\lambda = 0 \text{ or } 4$$

P.C (u) against eigenvalue 4 is

$$\begin{bmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2u_1 + 2u_2 = 0 \Rightarrow \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

P.C (u) against eigenvalue 0 is

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2u_1 + 2u_2 = 0$$

$$\text{thus } \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\text{Thus the PCA matrix } U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

(Q6)

Ans) Given :- $\{x_1, x_2, \dots, x_N\} \sim N(\mu, I)$ ~~Each~~ μ 's element is sampled from dist s.t

$$P(\mu) = \mathcal{Z} e^{-\lambda^T \mu}, \mu_i \geq 0$$

$$P(\mu) = 0, \mu_i < 0 \quad (\mu_i \text{ refers to } i^{\text{th}} \text{ element of } \mu)$$

R.T.F :- MAP estimate of μ

$$P(x_i | \mu) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)}$$

$$\begin{aligned} P(x | \mu) &= \prod_{i=1}^N \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)} \\ P(\{x_1, \dots, x_N\} | \mu) &= \prod_{i=1}^N \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)} \end{aligned}$$

$$P(x | \mu) = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)}$$

$$P(x | \mu) = (2\pi)^{-Nd/2} e^{-\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu)}$$

$$P(\mu | x) = \frac{P(x | \mu) \cdot P(\mu)}{P(x)}$$

$$P(\mu | x) = P(x | \mu) P(\mu) \quad (\text{discarding } P(x) \text{ in MAP estimation of } \mu)$$

$$= \left((2\pi)^{-\frac{Nd}{2}} e^{-\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu)} \right) P(\mu)$$

$$= (2\pi)^{-\frac{Nd}{2}} e^{-\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu)} \mathcal{Z}^d e^{-\lambda^T \mu}$$

$$p(\mu | x) = \left((2\pi)^{-\frac{Nd}{2}} \mathcal{Z}^d \right) e^{-\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu) - \lambda^T \mu}$$

Since $(2\pi)^{-\frac{Nd}{2}} \mathcal{Z}^d$ is a const., we can ignore it for $\max_{\mu} P(\mu | x)$.Moreover $\max_{\mu} \ln P(\mu | x)$ and $\max_{\mu} P(\mu | x)$. So, we need to minimize

$$\ln p(\mu | x) = -\frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu) - \lambda^T \mu$$

$$\equiv -\frac{1}{2} \sum_{i=1}^N (x_i^T x_i - x_i^T \mu - \mu^T x_i + \mu^T \mu) - \lambda^T \mu$$

$$\nabla_{\mu} \ln p(\mu | x) = -\frac{1}{2} \sum_{i=1}^N (-2x_i + 2\mu) - \lambda = 0$$

$$\Rightarrow \sum_{i=1}^N (x_i - \mu) - \lambda = 0 \Rightarrow N\mu = \sum_{i=1}^N x_i - \lambda$$

$$\hat{\mu}_{\max} = \frac{\sum_{i=1}^N x_i - \lambda}{N}$$

Q7)

Ans) Given :- $\max_w \frac{\text{tr}(W^T S W)}{\text{tr}(W^T W)}$

S is symmetric matrix

R.T.F :- W

$$W_{d \times c} = [w_1 \ w_2 \ \dots \ w_c]$$

$$S_{d \times d} = [s_1 \ s_2 \ \dots \ s_d]$$

$$W^T S W = [w_1 \ w_2 \ \dots \ w_c]^T [s_1 \ s_2 \ \dots \ s_d] [w_1 \ w_2 \ \dots \ w_c]$$

$$= \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_c^T \end{bmatrix}_{c \times d} \begin{bmatrix} s_1 & s_2 & \dots & s_d \end{bmatrix}_{d \times d} \begin{bmatrix} w_1 & w_2 & \dots & w_c \end{bmatrix}_{d \times c}$$

$$= \begin{bmatrix} w_1^T s_1 & w_1^T s_2 & \dots & w_1^T s_d \\ w_2^T s_1 & w_2^T s_2 & \dots & w_2^T s_d \\ \vdots & \vdots & \ddots & \vdots \\ w_c^T s_1 & w_c^T s_2 & \dots & w_c^T s_d \end{bmatrix}_{c \times c} \begin{bmatrix} w_1 & w_2 & \dots & w_c \end{bmatrix}_{d \times c}$$

$$= \begin{bmatrix} w_1^T S \\ w_2^T S \\ \vdots \\ w_c^T S \end{bmatrix}_{c \times d} \begin{bmatrix} w_1 & w_2 & \dots & w_c \end{bmatrix}_{d \times c} = \begin{bmatrix} w_1^T S w_1 & \dots & w_1^T S w_c \\ \vdots & \ddots & \vdots \\ w_c^T S w_1 & \dots & w_c^T S w_c \end{bmatrix}$$

Thus $\text{tr}(W^T S W) = \sum_{i=1}^c w_i^T S w_i$

Similarly $\text{tr}(W^T W) = \sum_{i=1}^c w_i^T w_i$

Hence $\max_w \frac{\text{tr}(W^T S W)}{\text{tr}(W^T W)} = \max_w \sum_{i=1}^c \left(\frac{w_i^T S w_i}{w_i^T w_i} \right) = \lambda_1$

let $y_i = \frac{w_i}{\sqrt{w_i^T w_i}} \Rightarrow y_i^T y_i = \frac{w_i^T w_i}{w_i^T w_i}$

$$\text{let } y_i = \frac{w_i}{\sqrt{w_i^T w_i}}$$

$$\text{thus } \max \sum_{i=1}^c \frac{w_i^T S w_i}{w_i^T w_i} \Rightarrow \max \sum_{i=1}^c y_i^T S y_i \quad \text{s.t. } y_i^T y_i = 1$$

Now apply langragian and find its maxima by apply grad wrt y_i

$$Z = \sum_{i=1}^c (y_i^T S y_i - \lambda (y_i^T y_i - 1))$$

$$\nabla_{y_i} Z = (2 S y_i - 2 \lambda y_i) = 0$$

$$\Rightarrow S y_i = \lambda y_i$$

$$\Rightarrow S y_i = \lambda y_i$$

$$\text{Remember } y_i = \frac{w_i}{\sqrt{w_i^T w_i}} \quad \text{Since } w = [w_1 \ w_2 \ \dots \ w_c]$$

$$\Rightarrow y = \begin{bmatrix} y_1 & y_2 & \dots & y_c \end{bmatrix} \quad \left(\text{as } \sqrt{w_i^T w_i} \text{ is a scalar and } \dim(y_i) = \dim(w_i) \Rightarrow \dim(y) = \dim(w) \right)$$

After applying with all the elements of y (i.e y_i where $i \in \{1, 2, \dots, c\}$)

$$\text{we get } S \begin{bmatrix} y_1 & y_2 & \dots & y_c \end{bmatrix} = \lambda \begin{bmatrix} y_1 & y_2 & \dots & y_c \end{bmatrix} \text{ or } S y = \lambda y.$$

Q8)

Ans)

$$\text{Given :- } X_1 = \begin{bmatrix} \overset{m_1}{2} & \overset{m_{12}}{0} \\ \underset{m_{21}}{0} & \underset{m_2}{2} \end{bmatrix}_{2 \times 2}$$

$$X_2 = \begin{bmatrix} \overset{m_1}{2} & \overset{m_{12}}{0} \\ \underset{m_{21}}{0} & \underset{m_2}{-2} \end{bmatrix}_{2 \times 2}$$

R.T.F :- FDA proj vector

$$\mu_1 = \frac{\sum_{i=1}^2 x_{1i}}{2} = \frac{x_{11} + x_{12}}{2} = \frac{\begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix}}{2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mu_2 = \frac{\sum_{i=1}^2 x_{2i}}{2} = \frac{x_{21} + x_{22}}{2} = \frac{\begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix}}{2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Sigma_1 = \frac{1}{N-1} (X_1 - \mu_1) (X_1 - \mu_1)^T = (X_1 - \mu_1) (X_1 - \mu_1)^T$$

$$= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

$$\Sigma_2 = \frac{1}{2-1} (x_2 - \mu_2)(x_2 - \mu_2)^T$$

$$= \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

In FDA we maximize the inter class distance

$$\max_u \|u^T(\mu_1 - \mu_2)\| \quad \text{or} \quad \max_u \|u^T(\mu_1 - \mu_2)\|^2 = \max_u u^T(\mu_1 - \mu_2)(\mu_1 - \mu_2)^T u$$

$$= \max_u u^T S_B u$$

and we minimize within the class variance/scatter

$$S_1 = N_1 \Sigma_1 = 2 \Sigma_1 = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}$$

$$S_2 = N_2 \Sigma_2 = 2 \Sigma_2 = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

Hence the projected, within the class, variance is

$$u^T(S_1 + S_2)u = u^T S_W u$$

we want to $\min_u u^T S_W u$

$$\text{Solve } \max \frac{u^T S_B u}{u^T S_W u}$$

$$\text{let } y = \frac{u}{\sqrt{u^T S_W u}} \Rightarrow \max_u \frac{u^T S_B u}{u^T S_W u} = \max_y y^T S_B y \quad \text{s.t. } y^T S_W y = 1$$

Thus, we want to apply Lagrange multiplier and solve for

$$\nabla_y (y^T S_B y - \lambda(y^T S_W y - 1)) = 0$$

$$2S_B y - 2\lambda S_W y = 0$$

$$S_B y - \lambda S_W y = 0$$

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$$

$$S_W = S_1 + S_2 = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$$

$$\left(\begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} \right) y = 0$$

$$\begin{bmatrix} -8\lambda & 0 \\ 0 & 4-8\lambda \end{bmatrix} y = 0$$

$$\begin{vmatrix} -8\lambda & 0 \\ 0 & 4-8\lambda \end{vmatrix} = 0$$

$$-8\lambda(4-8\lambda) = 0$$

$$\Rightarrow \lambda = 0 \text{ and } \lambda = \frac{1}{2}$$

eigenvector corr. to eigenvalue $\lambda = \frac{1}{2}$

$$\begin{bmatrix} -4 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$y_1 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

eigenvector corr. to eigenvalue $\lambda = 0$

$$\begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The FOA projection vector is $y_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$(Q9) \quad p(x|a,b) = \begin{cases} \frac{1}{b-a} ; a \leq x \leq b \\ 0 , \text{ otherwise} \end{cases}$$

$$LE(a,b) = \prod_{i=1}^n p(x_i|a,b) = \left(\frac{1}{b-a} \right)^n = \frac{1}{(b-a)^n} = (b-a)^{-n}$$

~~given~~ given $\forall i \in \{1, 2, \dots, n\}$ $x_i \in [a, b]$. Else $p(x_i|a,b) = 0$

For maximizing $LE(a,b)$, we need to minimize $b-a$.

$\Rightarrow b$ must be as small as possible & $b \geq x_i \quad \forall i \in \{1, 2, \dots, n\}$

$\Rightarrow a$ must be as large as possible & $a \leq x_i \quad \forall i \in \{1, 2, \dots, n\}$

Thus $\hat{a}_{MLE} = \min(D)$ & $\hat{b}_{MLE} = \max(D)$